

Machine Learning Model Outcomes

Executive summary report for TikTok prepared by the TikTok data team

Overview

The TikTok data team aims to develop a machine learning model to classify videos as either claims or opinions. Previous analysis of the available data indicated a strong correlation between video engagement levels and claim status. The team is confident that the final model will meet all performance expectations.

Problem

Due to the high volume of user reports on TikTok videos for various reasons, not all can be reviewed by human moderators. Videos containing claims, rather than opinions, are more likely to violate the platform's terms of service. To streamline the review process, TikTok aims to develop a method for identifying claim-based videos and prioritizing them for moderation.

Solution

To develop the final model, the data team built two tree-based classification models and evaluated their performance on a held-out validation dataset. The model with the highest recall score was selected and then applied to a test dataset to estimate its future performance.

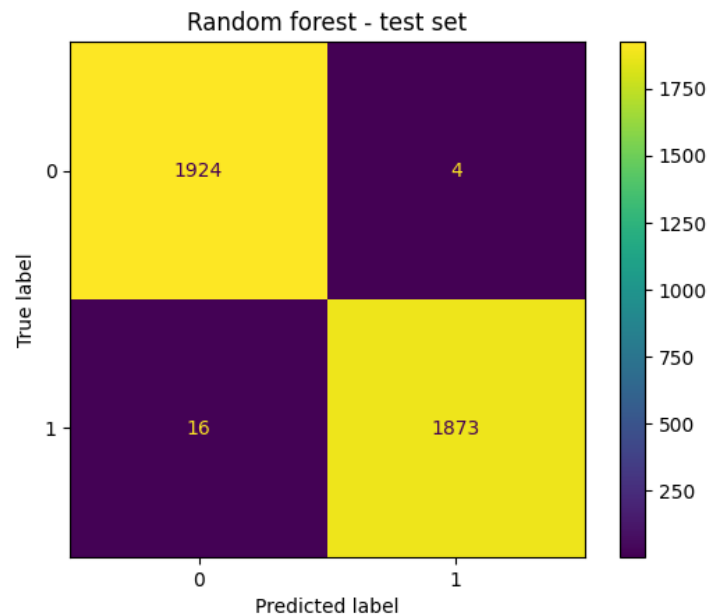
Details

The random forest (RF) and XGBoost model architectures both delivered outstanding performance. With a recall score of 0.999, the RF model outperformed the others and was chosen as the champion.

The model achieved near-perfect performance on the test holdout data, misclassifying only 20 samples out of 3,817.

The analysis revealed that, as anticipated, the key predictors were primarily tied to video engagement metrics. Specifically, video view count, like count, share count, and download count contributed almost entirely to the predictive signal in the data. Based on these findings, it can be concluded that videos with higher engagement levels were significantly more likely to be claims. Notably, no opinion video surpassed 10,000 views.

Confusion matrix for the champion RF model on test holdout data shows only five misclassified samples out of 3,817.



Next Steps

The model demonstrated excellent performance on the test holdout data. However, before deployment, the data team suggests conducting further evaluations using additional user data subsets. They also recommend monitoring the distributions of video engagement levels to ensure the model remains resilient to fluctuations in its key predictive features.